Slide-in motor supports with silent blocks — assembly report

Drill motor on a standing-desk crank axis (IKEA Trotten). Two supports slide onto four plots and lock with clip sample 4; TPU cushions and a TPU spider isolate the two vibration paths.

Principle

- Each support is a ring that clamps the motor plus a flat clip plate screwed on the ring. The plate has two U slots open toward the coupler.
- Four plots (Ø6 shoulders) are screwed into the plank, two per support, at y = −37 and +19 (pitch 56 mm). Sliding the motor along its axis engages the plots in the slots; the jaws of the coupling engage at the same time.
- Two printed tongues at the mouth of each slot spread by 0.30 mm, then close behind the plot (clip sample 4). The closed end of the slot is the forward stop. Nothing to tighten after sliding.

Vibration isolation: two paths, one elastic element each

- Mount path (drill body → wood): on every plot the plate floats between two TPU cushions (8 in total) with 0.3 mm of play on each side. No preload, so no friction when sliding in; the torque and the weight always meet TPU, never PETG on wood.
- Shaft path (drill → desk axis): the jaw coupling with its TPU spider (pages 14–16), carried over from the previous prototype. Hub A and the drill adapter are now one printed part, the drill hub (page 16).
- Estimate: cushion ≈ 611 N/mm, natural frequency ≈ 454 Hz, isolation above ≈ 641 Hz (page 8). To be measured with the ADXL345.

What changes compared with the previous prototype

- Removed: ears with bobbin silent blocks (they broke at the Ø9 hole), blades with TPU stop patches, one wood screw per mount.
- The silent blocks move from the ears to the plots: flat cushions, no neck, no small feature. The ring supports keep your rings and detrompeurs and carry a pad instead of ear and blade.

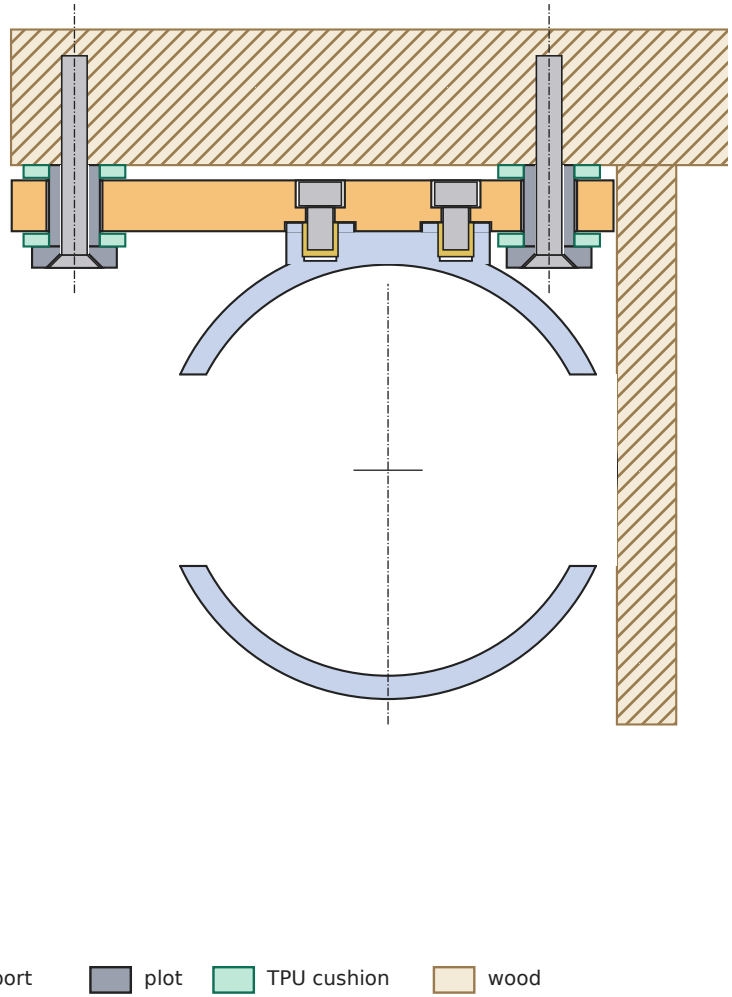
Desk constraints used

- Plank underside 36 above the axis; vertical wood face 27 from the axis; plank edge −44.48 on the ear side (motor_mount).
- Stack: plank 36.0 · cushion 1.5 · play 0.3 · plate 34.2 → 28.2 · play 0.3 · cushion 1.5 · head Ø10 × 2.5 (bottom at z = 23.9).

To validate

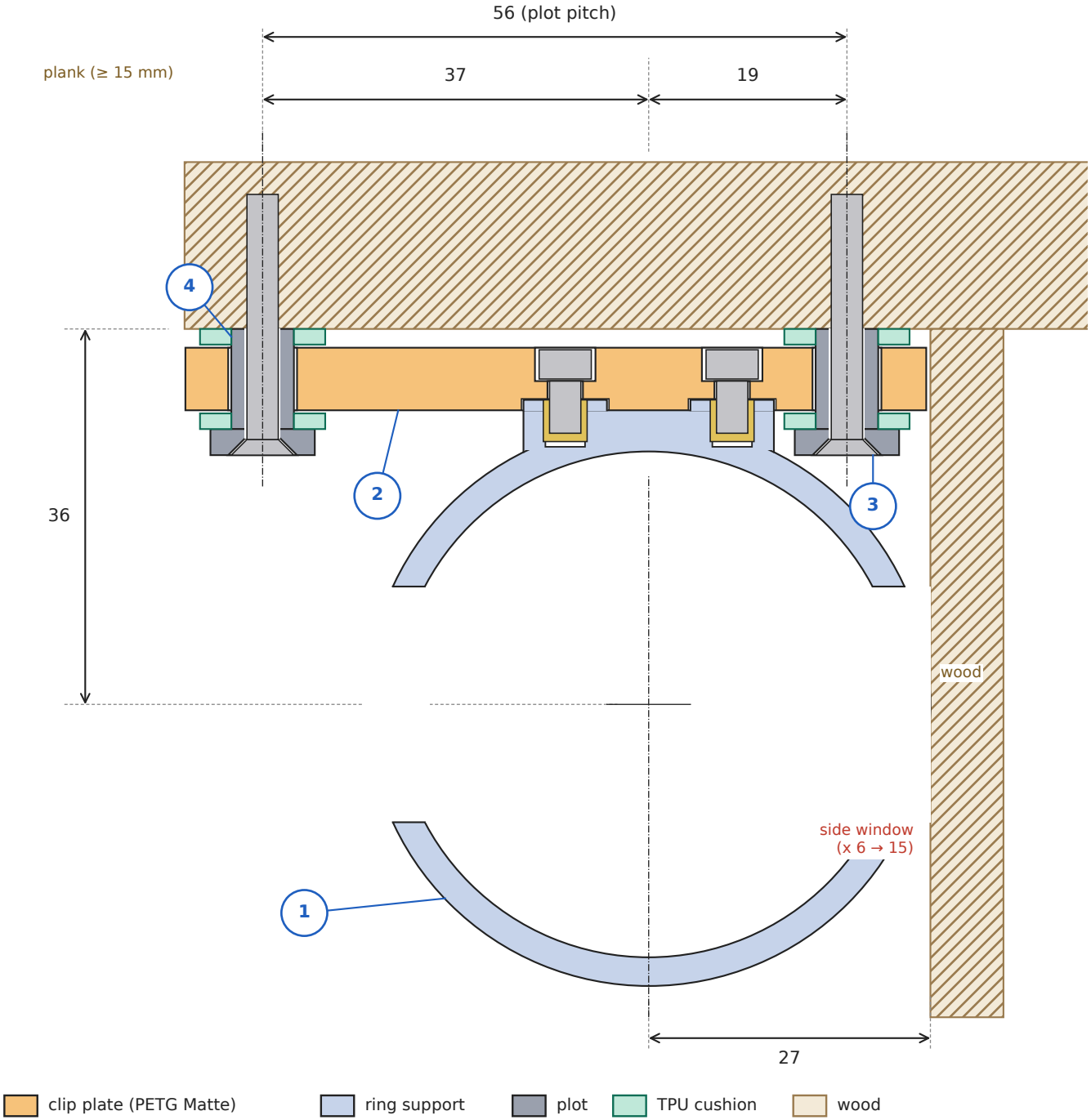
- Slot mouths toward +x = toward the coupler (assumed). Rings and detrompeurs are taken from your two SLDPRT (checked to 0.002 mm); only their direction along x is assumed.
- Real distance from the shaft axis to the plank underside (36.0 assumed): it sets the cushion thicknesses (page 7).

Section through the plot axes (rear support)



Assembly — overview

Section y–z through the plot axes, motor axis = x (perpendicular to the page, toward the coupler). Rear support (motor_mount ring Ø48.5) shown. Dimensions in mm.



Parts

1 Ring support, one per motor ring (×2)

2 Clip plate, two slots, 2 × M3 (×2)

3 Plot with its Ø3 × 25 wood screw (×4)

4 TPU cushion, 2 per plot (×8)

Assembly order (pictures page 12)

1 Heat-set inserts. Press 2 M3 × 4 inserts into the bosses of each ring support pad (iron ≈ 230 °C), flush with the boss face.

2 Plates. Screw each plate on its pad with 2 × M3 × 5, mouths toward the coupler (+x). ≤ 0.3 N·m.

3 Cushions. Slide 2 cushions on the shoulder of each plot: one down to the head, one up to the flat top.

4 Plots. Drill the plank (page 11) and screw the 4 plots, flat top against the plank, at y = −37 and +19.

5 Slide. With the motor in its two rings, slide the whole assembly toward the coupler: each plot spreads the tongues, the clip snaps behind it, the slot end stops the motor, the coupling jaws engage.

6 Check. Turn the axis by hand: no tight spot, plate free between the cushions (0.3 mm each side). Axis offset ≤ 0.3 mm, tilt ≤ 1°.

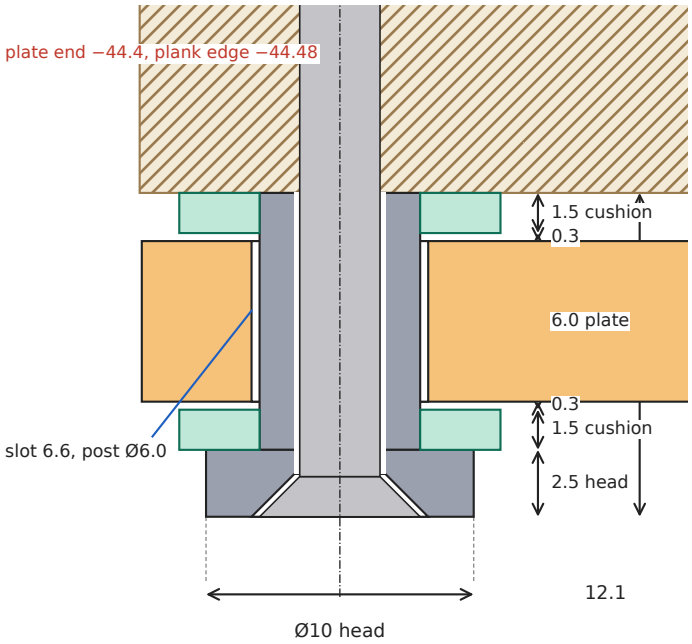
7 Remove. Pull the motor back: the four clips release.

Dimensions: SolidWorks sketches (plank 36, wood 27) and build_slide_mount.py.

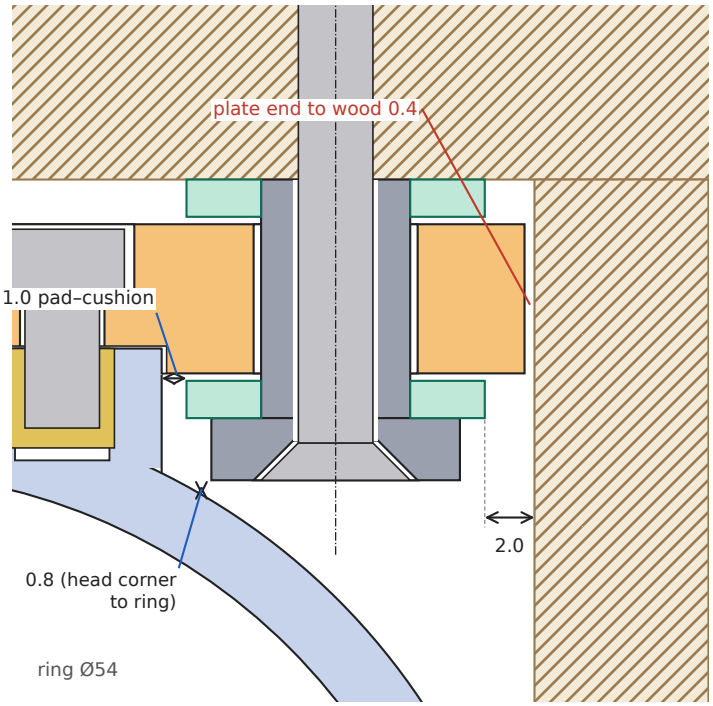
Assembly — stack and clearances (dimensioned details)

Three zoomed sections of the same drawing. z up, y toward the wood. Dimensions in mm.

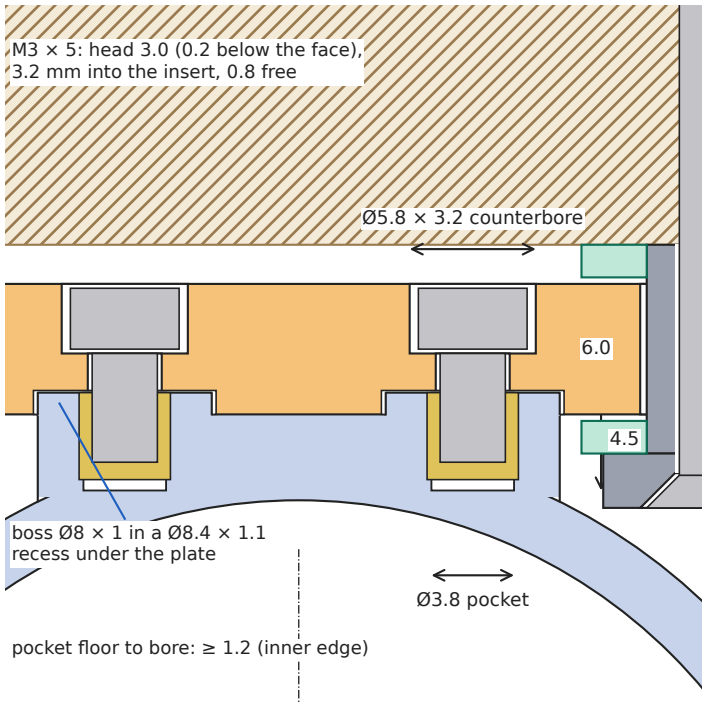
A — plot on the ear side (y = −37)



B — plot on the wood side (y = +19)



C — plate on the pad (screw at y = +8)



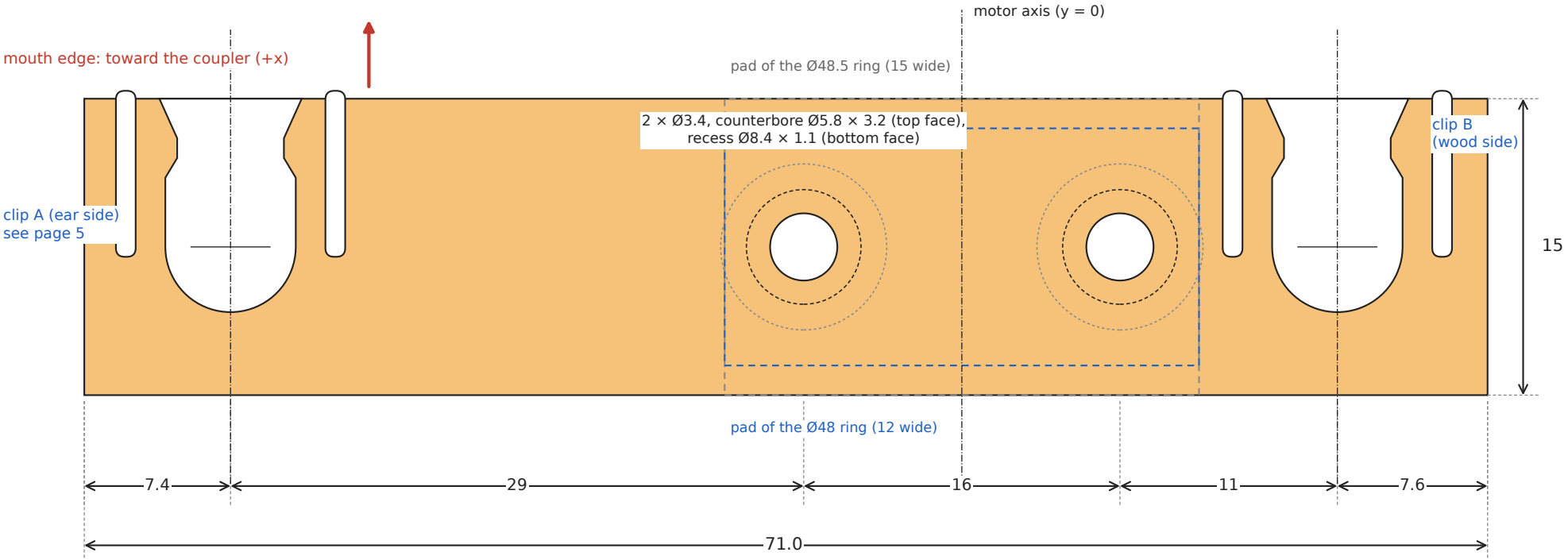
Item	Value	Comment
plank underside	z = 36.0	from the SolidWorks sketch
top cushion	36.0 → 34.5	against the plank, friction fit on the shoulder
plate top / bottom	34.2 / 28.2	0.3 play above and below, no preload
bottom cushion	27.9 → 26.4	on the head, friction fit on the shoulder
plot shoulder / head	Ø6.0 × 9.6 / Ø10 × 2.5	head bottom at z = 23.9
pad top = plate bearing face	z = 28.2	1.2 above the ring top (27.0)
plate end (ear / wood side)	y = −44.4 / +26.6	plank edge −44.48; wood face 27
ring to wood	1.0	relief flat at y = 26, |z| < 7.4
head B to ring	0.8	closest point, ring Ø54 at y = 14

Notes

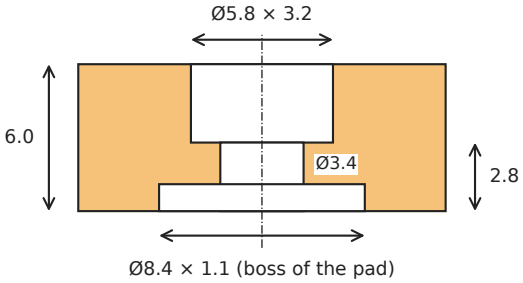
- The plate is carried by the pad and floats between the cushions (0.3 mm of play each side): the plots only guide it, hold it down and take the couple, always through TPU.
- Plot B is the tight one: 2.0 to the wood, 1.0 to the pad, 0.8 to the ring. If the wood face is not at 27 from the axis, move plot B (the slot has no adjustment).
- The plank edge on the ear side limits plot A and the plate end (motor_mount side −44.48; front mount side −48.09).

Clip plate — drawing (×2, identical)

Plan view from the plank side, motor axis x vertical (mouth edge at the top, toward the coupler), y horizontal. Printed flat, bottom face (on the pad) on the bed. Dimensions in mm.



Section through a screw (y = ±8)

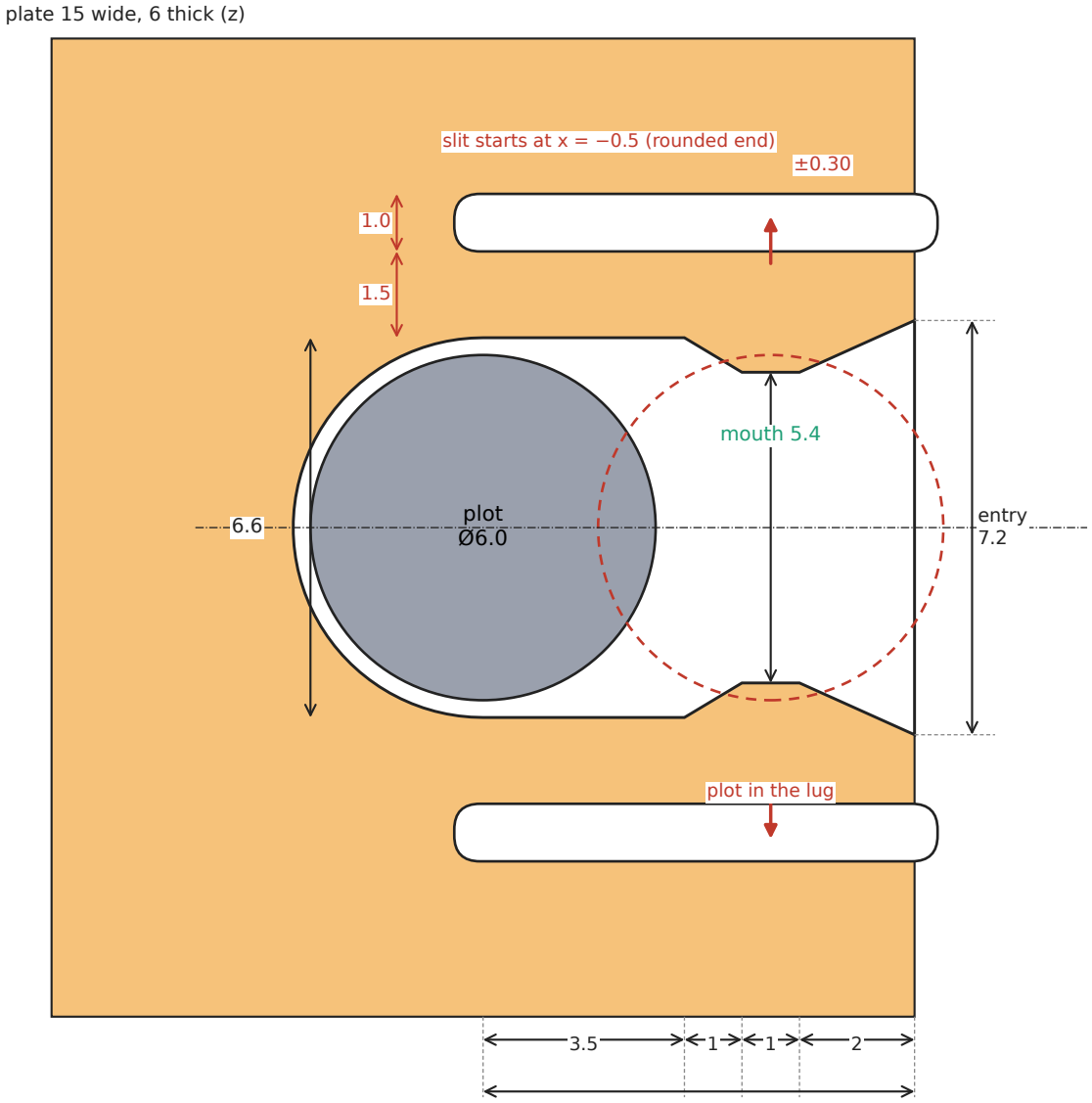


Notes

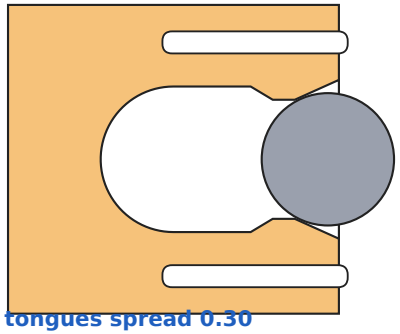
- Two identical plates, one per ring support. The two slots are the clip of sample 4 (page 5), pitch 56, centred on y = −37 and y = +19.
- The plate is fixed to the pad by 2 × M3 × 5 only; the two bosses of the pad fit in its recesses (0.2 mm play) and centre it on the ring in x (15 mm plate on a 12 mm ring: 1.5 mm overhang each side).
- Print flat so that the tongues bend in the plane of the layers: 5 walls, no support, bottom face on the bed.
- If your coupler is on the other side of the motor, use the mirrored plate (same part, mouths toward −x).

Snap clip — sample 4 (dimensioned) and how it works

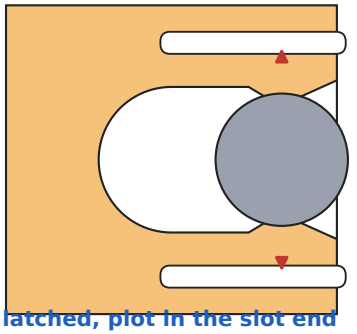
Plan view, origin on the plot axis, mouth toward +x. Tongue 1.5 mm, interference 0.30 mm per side, mouth 5.4 mm for a Ø6.0 plot. Dimensions in mm.



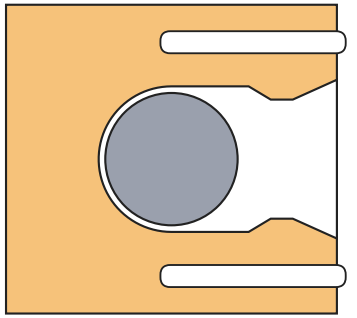
1 entry



2 tongues spread 0.30



3 latched, plot in the slot end



Numbers (beam theory, to be measured)

Tongue = cantilever 5 mm long, 1.5 thick, 6 high, flexural modulus 2 GPa, friction 0.3.

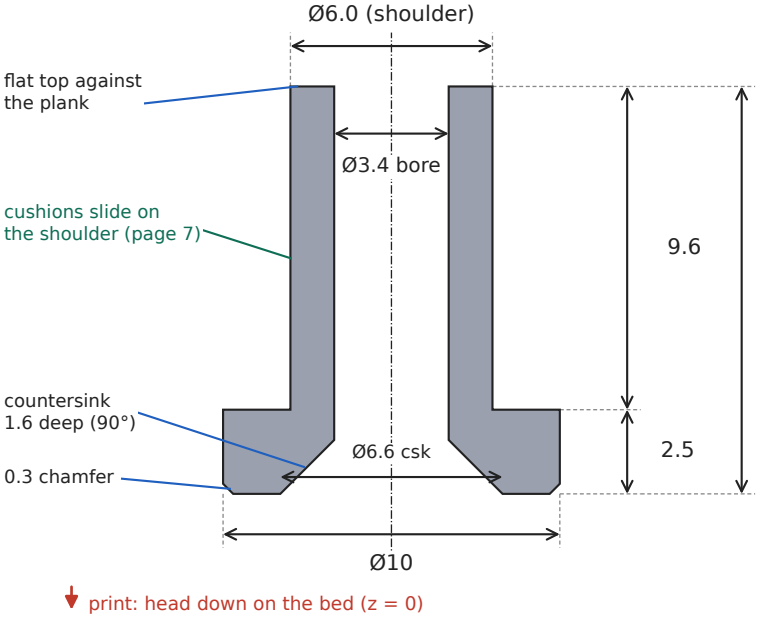
Push-through: tongue force ≈ 24 N each; unlatching ≈ 50 N per clip (±50 %).

Strain at the tongue root ≈ 2.7 %: close to the PETG Matte limit in the layer plane (≈ 3 %), above it across layers (≈ 1 %), hence the flat print.

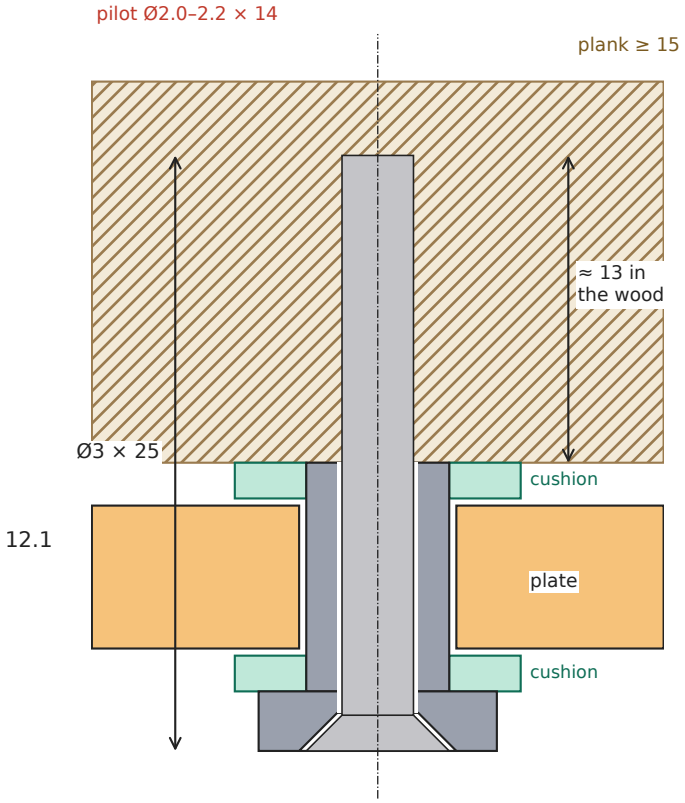
Four clips in parallel: pulling the motor out ≈ 210 N. Measure; if too stiff use sample 3 (page 17).

Plot — part drawing and installation (×4)

Printed PETG Matte, head on the bed, no support. Axial section, z up. Dimensions in mm.



Installed (plot A shown)



Fit and print

- The shoulder diameter sets the clip interference (0.30 per side at Ø6.0) and the grip of the cushions: measure the printed plot with calipers, 6.0 ± 0.05.
- Head down, no support: the countersink is a 45° cone, the 0.3 chamfer takes the first-layer squash.
- 4 walls: the 1.3 mm wall around the bore is solid. The plot is lightly loaded: ≈ 30 N axial (the couple) and a few N sideways, ≈ 2 MPa across the layers at the head (PETG Matte: 30 MPa).

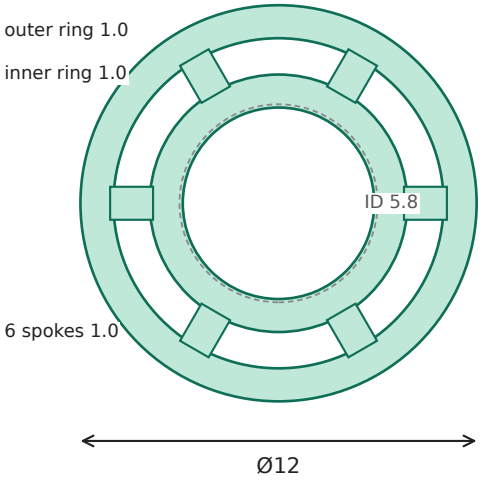
Hardware

- Wood screw Ø3 × 25, countersunk head Ø6, pilot Ø2.0–2.2, ≈ 13 mm in the wood. With a plank thinner than 15 mm use Ø3 × 20.
- Alternative: M5 shoulder screw (shoulder Ø6 × 10) in an M5 wood insert, with the same two cushions.

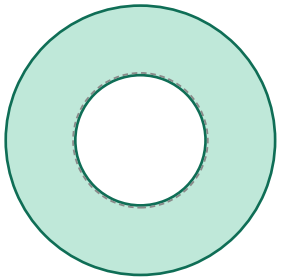
Silent blocks — TPU cushions (×8) around the plate on each plot

Flat ring washers printed in TPU 85A, 0° overhang, no support, no small feature. Two per plot, friction fit on the Ø6.0 shoulder. Dimensions in mm.

wheel (default)

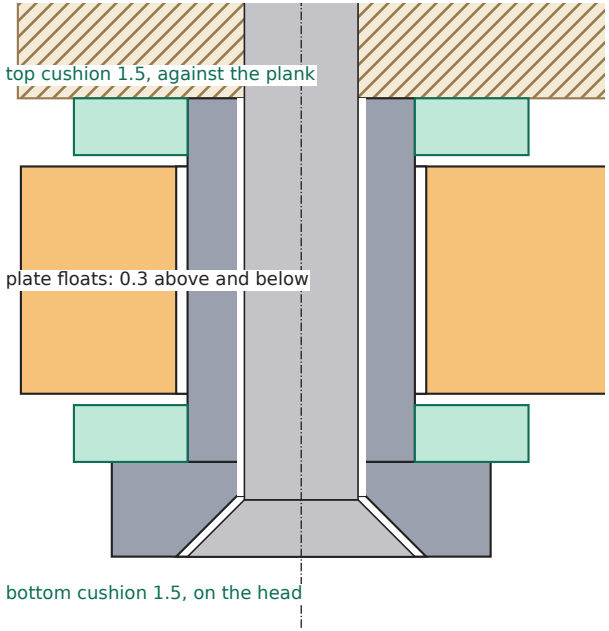


solid (stiff variant)



thickness 1.5, both variants

Installed, section through plot A



What it does

- The plate sits between two cushions with 0.3 mm of play on each side: nothing is pinched, so it slides in without friction and the clip is not disturbed.
- Under torque or weight the plate rests on a cushion, never on PETG or wood. The play is a few tenths of a degree of rotation.
- Both directions of rotation are covered: the post that is pushed up meets the top cushion, the one pushed down meets the bottom cushion.

Variant	Stiffness*	Natural freq.*	Use
wheel (default)	≈ 611 N/mm	≈ 454 Hz	filters from ≈ 650 Hz
solid	≈ 2174 N/mm	≈ 857 Hz	resonates near 1 kHz: avoid
wheel, 2 walls	softer	lower	first thing to try

* TPU 85A E ≈ 12 MPa (assumed), 4 bottom cushions in parallel, mass 0.3 kg; page 8. Measure before trusting.

Print and fit

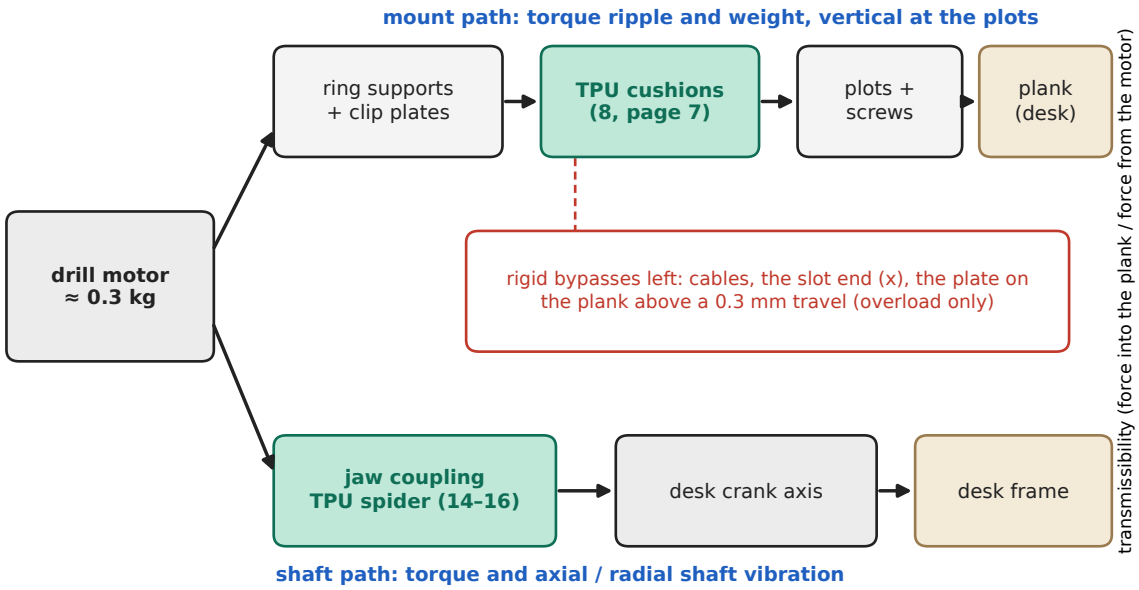
- TPU 85A on the external spool, dried (70 °C, 8 h), 225 °C, bed 35 °C, 20 mm/s, fan 100 %, textured plate without glue (settings page 13). 2 walls, thin-wall detection on, 3 top and bottom layers.
- The inner ring is ×5.8 on a Ø6.0 shoulder: a firm friction fit (0.1 mm per side) keeps each cushion where it is put. TPU printed undersize needs no more; if it slips, print the ID at 5.7.
- Stiffness tuning: print 3 sets at 2 / 3 / 4 walls, measure on the plank with the ADXL345, keep the softest set that does not let the motor move under load.

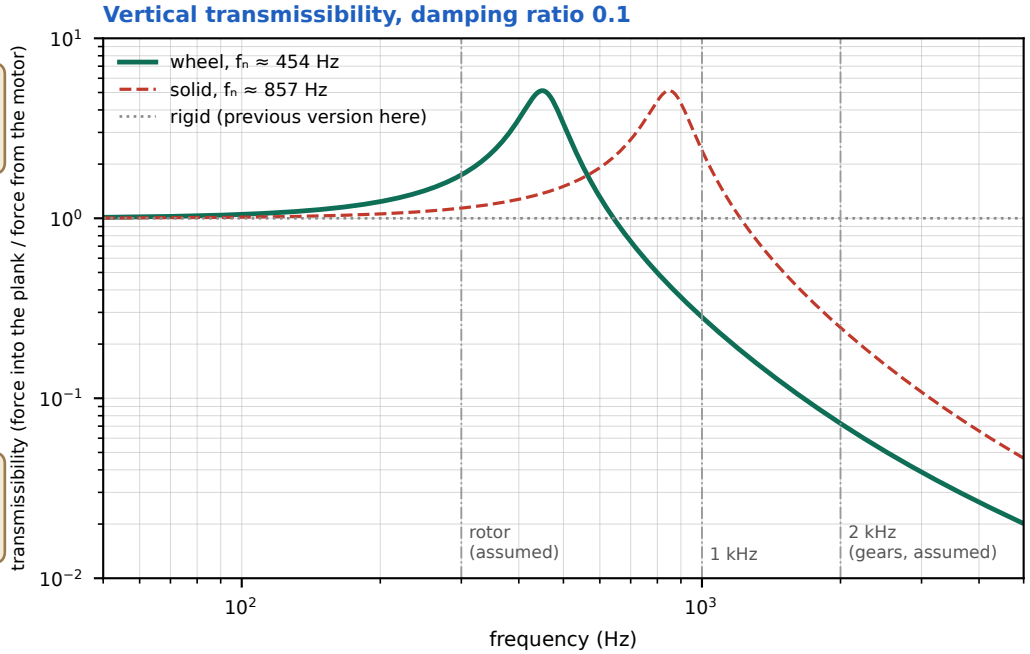
Set the thickness for your desk

- The plate is fixed to the ring, the plots hang from the plank: the two plays depend on H, the real distance from the shaft axis to the plank underside (36.0 assumed).
- With d = H − 36: top cushion 1.5 + d, bottom cushion 1.5 − d keep 0.3 mm of play on both sides (layers of 0.2). Or set PLANK_Z in build_slide_mount.py: the plate height, the pad and the plots are recomputed.
- Measure H on the desk before printing the ring supports and the plots.

Vibration paths and isolation estimate

Two elastic elements, one per path. The numbers are estimates from assumed TPU properties: they tell what to expect, the ADXL345 tells what you get.





Quantity (estimate)	wheel	solid
cushion stiffness	611 N/mm	2174 N/mm
natural frequency (4 bottom cushions)	454 Hz	857 Hz
transmissibility at 1 kHz	0.28 (-11 dB)	2.4 (resonance)
transmissibility at 2 kHz	0.072 (-23 dB)	0.25 (-12 dB)
deflection under the nominal couple	0.05 mm	0.01 mm
rotation to take up the plays	0.9° at B, 0.5° at A	same

Assumptions

TPU 85A printed E ≈ 12 MPa; damping ratio 0.10; moving mass 0.3 kg (motor 0.2–0.3 kg from your note, plus two supports); rotor and gear frequencies are typical values: replace them by your own spectrum.

Measure it

- ADXL345 on the plank near plot A and on the ring support, same axis, same PWM. Record 10 s at three PWM values. The ratio of the two spectra is the transmissibility: compare with this curve and with the rigid version (cushions removed, plate resting on the heads).

Reading

- The wheel isolates above about 641 Hz: gear whine and PWM harmonics, not the rotor frequency. For lower frequencies the cushion would have to be much softer, which the 3.3 N·m torque and the 0.3 mm alignment tolerance do not allow.
- The solid washer is stiffer and sits on a resonance near 1 kHz: keep it as a spare, not as the default.

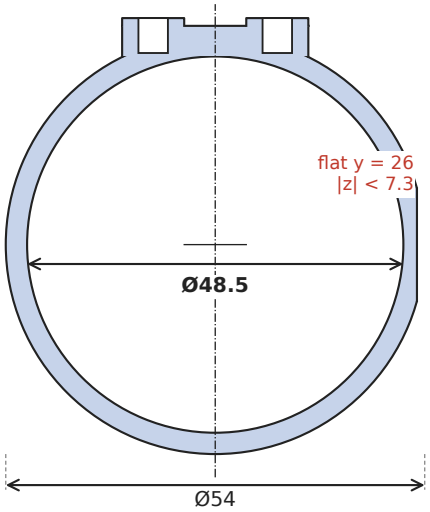
Torque split between the two plots of a support

- Nominal couple per support 1.65 N·m: 29 N if the posts shared it equally. With equal plays and cushions, post A (lever 37) meets its cushion first: 36 N at A, 17 N at B, and the 19 N difference goes radially through the coupling (assumed 200 N/mm): axis shift ≈ 0.09 mm, inside the 0.3 mm the coupling accepts.
- Smaller plays reduce it (GAP in the script); matching the plays to the lever arms (about 2:1 between A and B) nearly removes it. Check the axis offset on the desk (page 18).

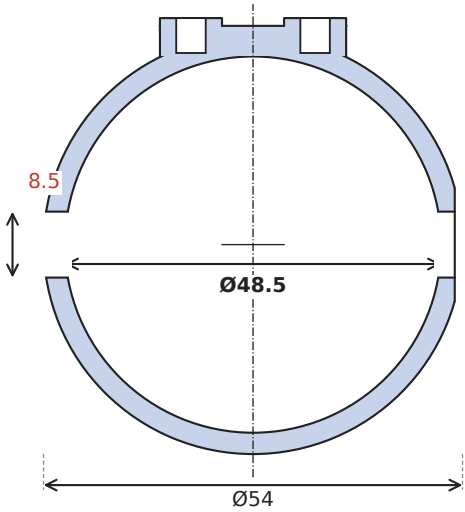
Ring support motor_mount (×1) — geometry of your original

Ring, side slots and windows taken from motor_mount_1.SLDPRT: every original vertex lies within 0.002 mm of this model. Ear, blade and gussets give way to the pad. Dimensions in mm.

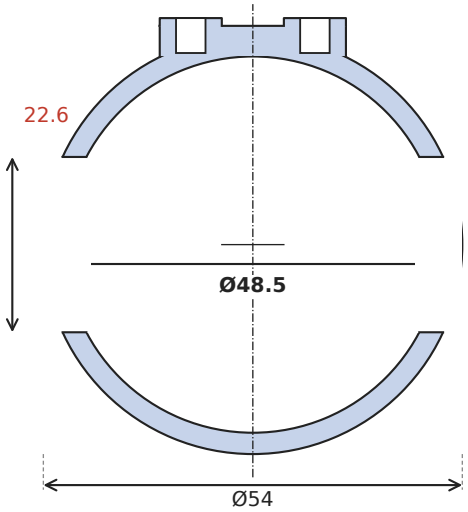
A–A x = 0 → 3 (closed rim)



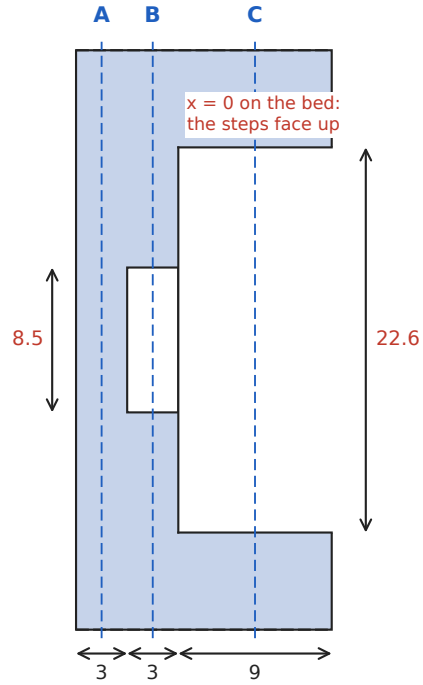
B–B x = 3 → 6 (side slots)



C–C x = 6 → 15 (side windows)



Side view from +y (side wall, |z| < 17)



Feature	Value	Read in
bore / outside, width	Ø48.5 / Ø54, 15	cylinders R24.25, R27, x 0 → 15
side slot, both sides	x 3 → 6, |z| < 4.25	planes x = 3, x = 6, z = ±4.25
side window, both sides	x 6 → 15, |z| < 11.3	planes x = 6, z = ±11.3
relief flat on the wood side	y = 26, |z| < 7.3, x 0 → 6	plane y = 26 (your 45° chamfer up to the blade is not kept)
pad, bosses (new)	24 wide, top z = 28.2	bosses Ø8 × 1, pockets Ø3.8 × 4.5 at y = ±8

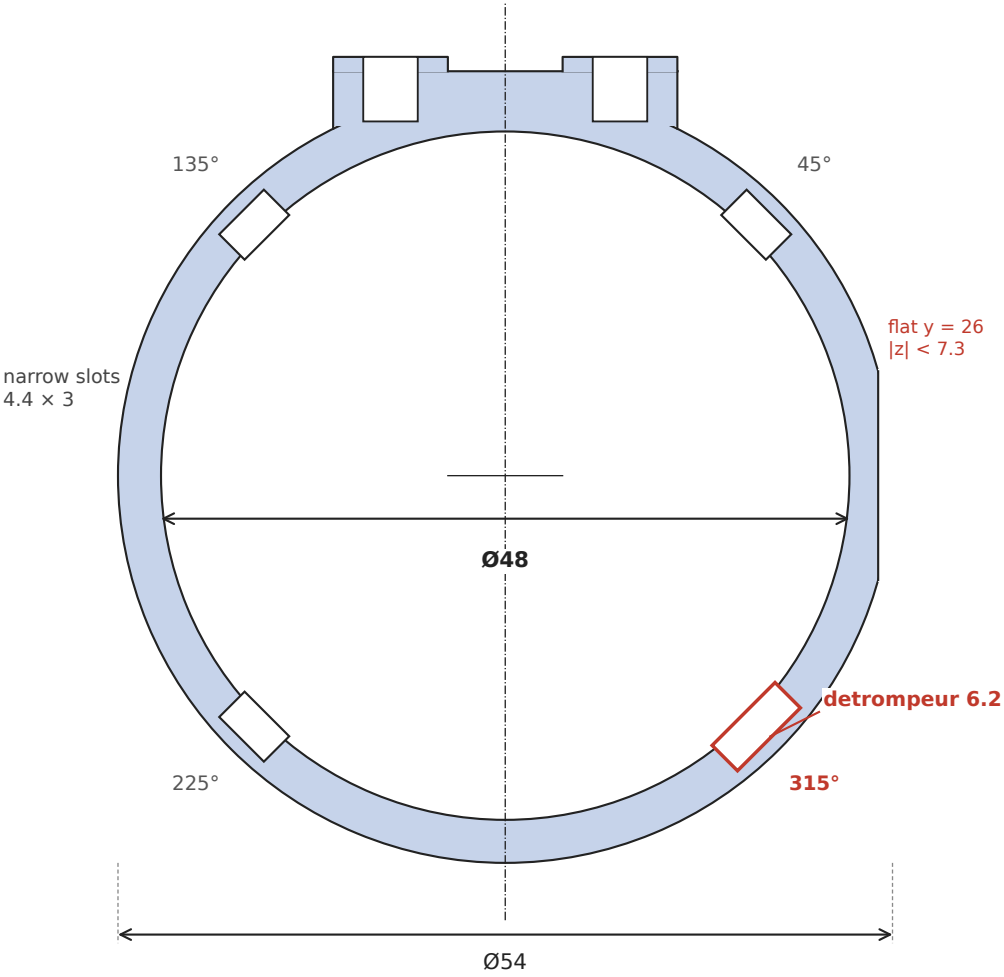
What these steps are

- They are the detrompeurs of the drill body: the ring is closed for the first 3 mm, narrows to side slots for 3 mm, then opens into two side windows. The drill cannot be fitted the wrong way round and the windows take its torque.
- Printed with the x = 0 rim on the bed (closed ring down), the steps face upward and the windows open upward: no support, nothing to bridge. Same orientation as the file.
- Behind the rim the ring is two arcs: the pad sits on the upper one, carried by the 6 mm rim. 4 walls, and look at the first print for flex under the couple.
- Not reproduced: ear, blade, gussets and fillets of your original, replaced by the pad.

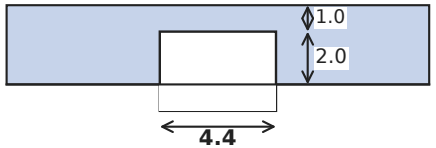
Ring support motor_front_mount (×1) — four slots, one of them the detrompeur

From motor_front_mount_1.SLDPRT, within 0.002 mm. Desk frame (wood on +y): your file is mirrored, so its wide slot at 225° is at 315° here (bottom right), as you confirmed.

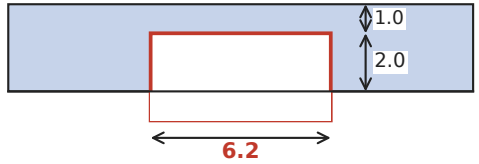
End view (desk frame)



narrow slot (45°, 135°, 225°)



wide slot (315°): the key



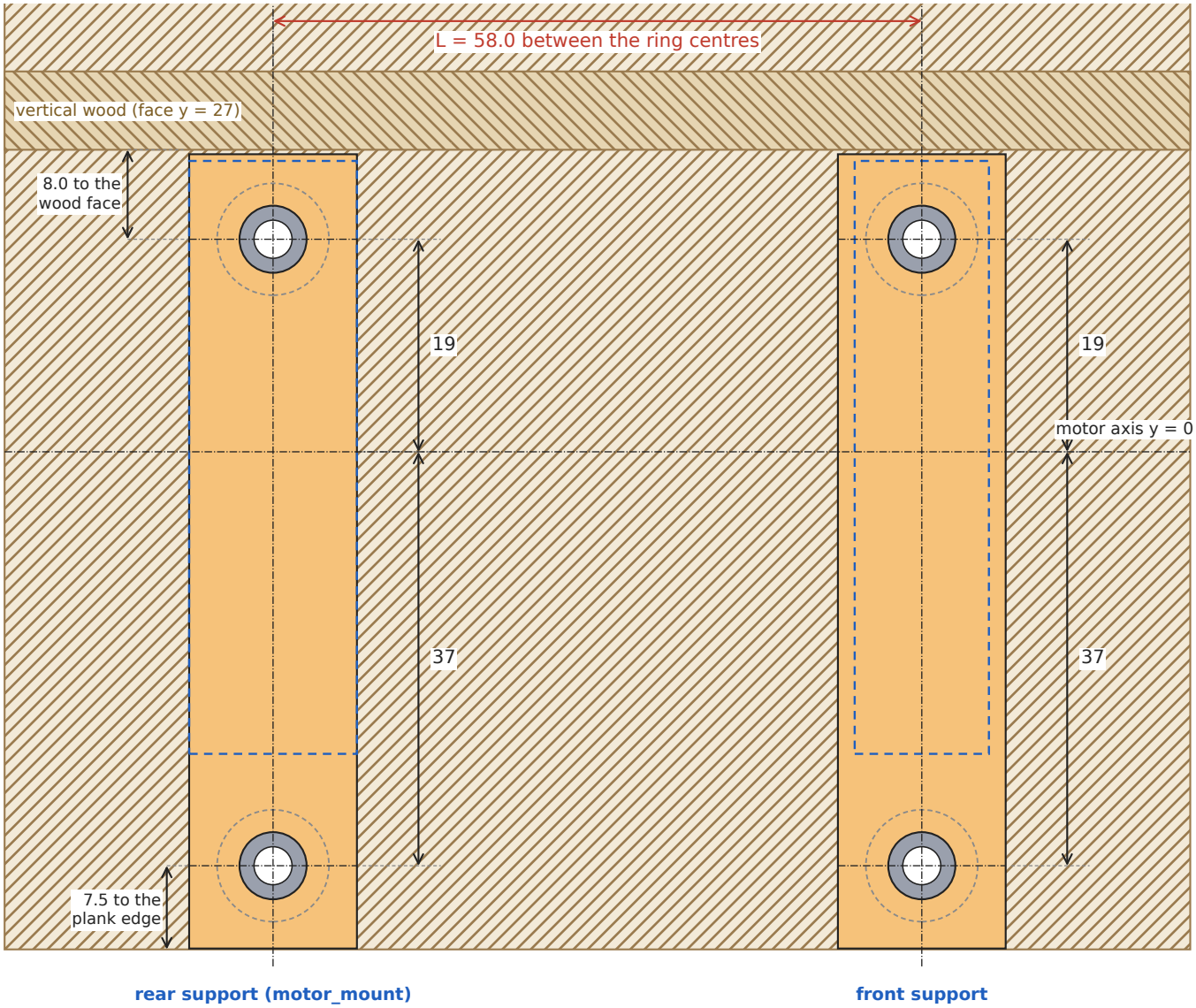
Feature	Value	Read in
bore / outside, width	Ø48 / Ø54, 12	cylinders R24, R27, x 0 → 12
slots, flat bottom	26.0 from the axis	2.0 beyond the bore, on the diagonals
narrow slots (×3)	4.4 wide, at 45°, 135°, 225°	planes at ±2.2 from the diagonal
wide slot, the key (×1)	6.2 wide, at 315° (bottom right)	planes at ±3.1 from the diagonal
relief flat on the wood side	y = 26, |z| < 7.3, full width	plane y = 26
pad, bosses (new)	24 wide, top z = 28.2	bosses Ø8 × 1, pockets Ø3.8 × 4.5 at y =

Orientation and print

- The slots are prismatic along x and the ring is symmetric end to end: either face can go on the bed. 4 walls, no support.
- The key is at the bottom right on this view (coupler side, wood on the right): your file has it at 225° in its own frame (wood on the left), which becomes 315° here. FRONT_KEY_ANGLE in the script moves it.
- Width 12, as in your file (the ear widening to 15 was not applied there); the plate overhangs the ring by 1.5 mm each side.

Plank layout — drilling points for the four plots

View from below (plank underside), motor axis x horizontal toward the coupler. L = 44.5 between the rings (measured) + 7.5 + 6 = 58.0. Dimensions in mm.



Point	x	y	Drill
P1 rear, ear side	0	−37.0	Ø2.0–2.2 × 14
P2 rear, wood side	0	+19.0	Ø2.0–2.2 × 14
P3 front, ear side	58.0	−37.0	Ø2.0–2.2 × 14
P4 front, wood side	58.0	+19.0	Ø2.0–2.2 × 14

x = 0 on the rear ring centre, y from the motor axis (+ toward the wood). Tolerance of the pitch between P1 and P2: ±0.3 (the slots have 0.3 mm play per side).

Drilling jig (stl/drill_jig.stl, ≈ 3 g)

- A light frame with a guide on each plot and two lips that bear on the vertical wood face. Print it flat, the plank face on the bed, no support; ream each guide once with the drill bit.
- Plates on the rings, coupling fully engaged (motor pushed toward the coupler): mark the two edges of each plate on the vertical wood face. Hold the jig flat on the plank, lips on the wood, lip ends on the marks. Both lips must match the marks: that checks L.
- Depth stop at 14 + 8 (guide) = 22 mm from the tip of the bit.

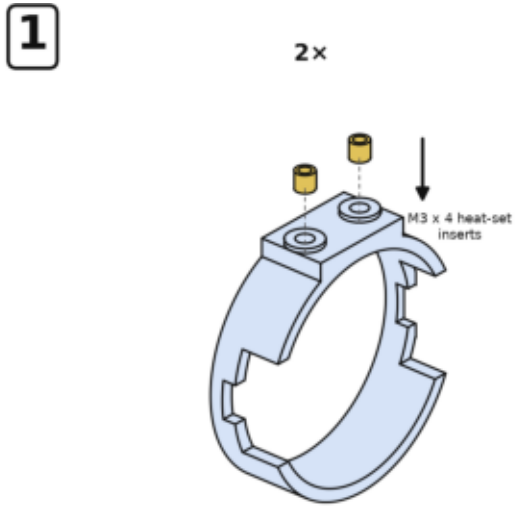
Drilling

- Pilot Ø2.0–2.2 for a Ø3 wood screw, 14 mm deep (the screw is 25 mm, the plot is 12.1 mm long, so ≈ 13 mm enters the wood). Check the plank thickness first.
- Screw the plots with the flat top fully against the plank. Do not over-tighten: the plot is printed.

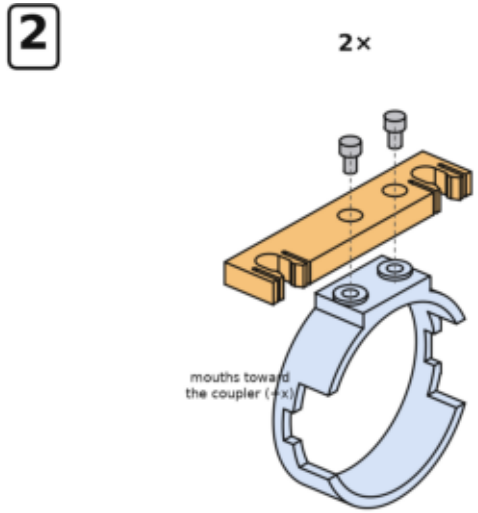
clip plate · ring footprint (dashed) · plot · plank

Assembly sequence

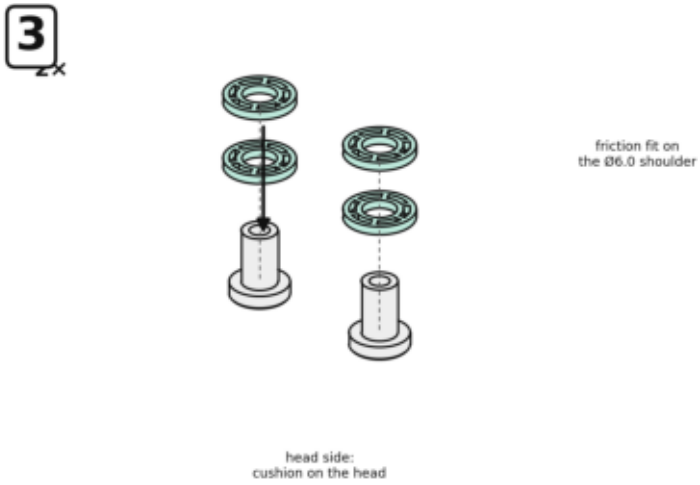
IKEA-style pictures generated from the CAD (docs/img). Plate = orange, ring support = blue, plot = grey, TPU cushion = green, wood = plank.



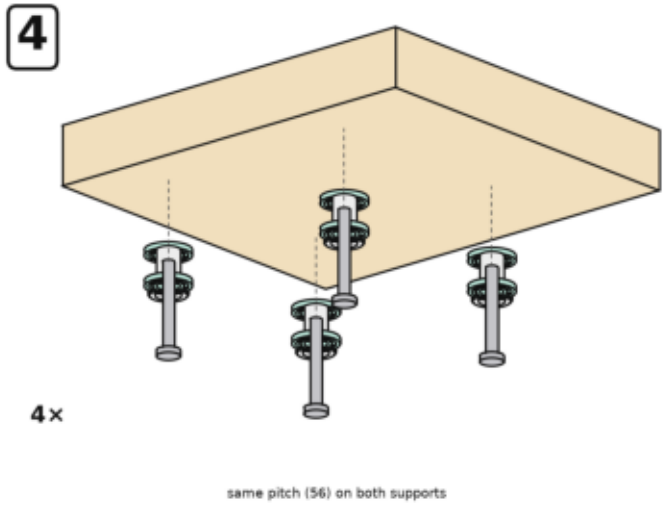
1 Press 2 M3 × 4 heat-set inserts into the bosses of each pad (iron ≈ 230 °C), flush with the boss face.



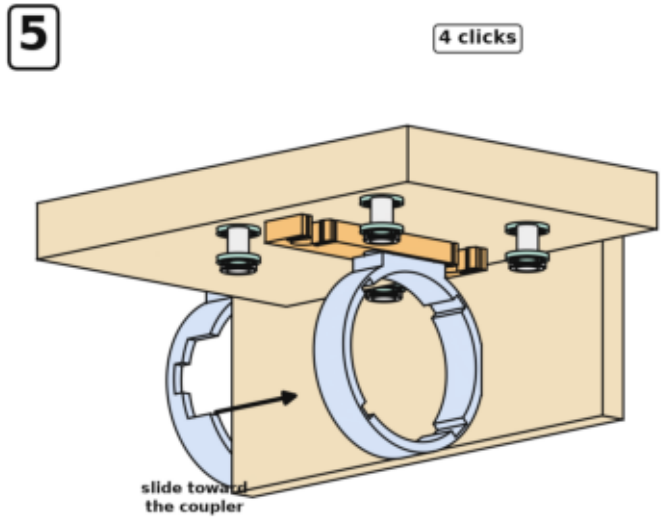
2 Screw the plate on the pad (the bosses centre it): 2 × M3 × 5, mouths toward the coupler (+x). ≤ 0.3 N·m.



3 Slide 2 cushions on every plot: one down to the head, one up to the flat top. Friction fit.



4 Pilot Ø2.0–2.2 × 14, then screw the 4 plots (Ø3 × 25) with the flat top fully against the plank.



5 Slide the motor (rings and plates on) toward the coupler: the four clips click, the slot ends stop it.

Check

Turn the axis by hand: no tight spot. The plates must be free on the plots (0.3 mm above and below, no preload). Axis offset ≤ 0.3 mm, tilt ≤ 1° at the coupling.

Remove

Pull the motor back along its axis: the four clips release one after the other.

Coupling

Order of the jaw coupling (hubs, spider): pages 14 to 16.

Print settings and bill of materials

Bambu Lab A1 mini, 0.4 mm nozzle, Bambu Studio. Dry the filaments before printing. Files in stl/ (step/ for CAD), the drill hub in solidworks/.

Printed parts

ID	Part	Qty	Material	Orientation	Walls / infill	File
A	Clip plate	2	PETG Matte	flat, bottom face on the bed	5 walls, 25 %	clip_plates_x2.stl
B	Plot	4	PETG Matte	head on the bed	4 walls (solid)	plots_x4.stl
C	TPU cushion, wheel (silent block)	8	TPU 85A	flat	2 walls, 3 top/bottom	cushions_x8.stl
D	Ring support, motor_mount (Ø48.5 × 15)	1	PETG	ring axis vertical	4 walls, 25 %	ring_support_motor_mount_print.stl
E	Ring support, front mount (Ø48 × 12)	1	PETG	ring axis vertical	4 walls, 25 %	ring_support_front_mount_print.stl
F	Drill hub (drill adapter V4 + hub A)	1	PETG	threaded end on the bed	4 perimeters	DrillAdapter+hub.STL
G	Coupling hub B (flange side)	1	PETG	base on the bed	4 perimeters	hub_b_flange_side.stl
H	Spider (TPU, coupling)	1 + 1	TPU 85A	flat, lobes upright	100 %, 4 walls	spider_tpu.stl
I	Drilling jig (tool, page 11)	1	PETG	plank face on the bed	all walls	drill_jig.stl
–	optional: plate mirrored, solid cushion	–	as A / C	as A / C	as A / C	clip_plate_mirrored.stl, cushion_solid.stl

Hardware

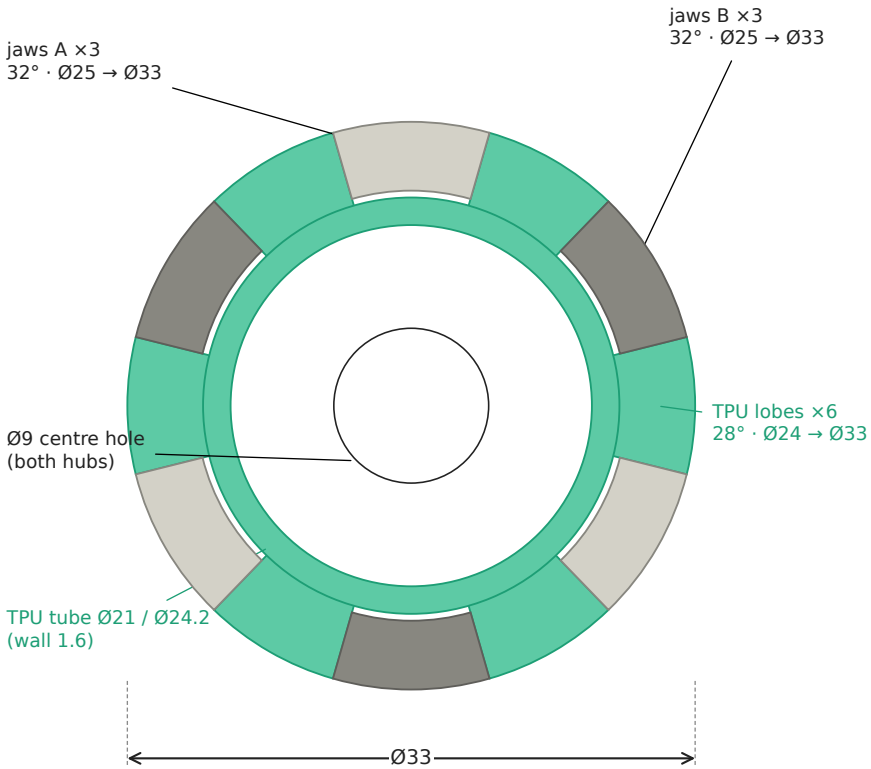
ID	Item	Qty	Used for
1	Wood screw Ø3 × 25, countersunk head Ø6	4	plots into the plank, pilot Ø2.0–2.2 × 14, ≈ 13 mm in the wood
2	M3 heat-set insert M3 × 4 × Ø4.2 (pocket Ø3.8 × 4.5)	4	ring supports, 2 per pad, in the bosses
3	M3 × 5 cap screw, head Ø5 × 3	4	plates on the pads (counterbore Ø5.8 × 3.2)
4	M3 heat-set insert M3 × 4 × Ø4.2 (pocket Ø3.8 through)	4	hub B
5	M3 × 6 cap screw, head Ø5 × 3	4	flange coupler onto hub B
6	Rigid flange coupler, bore 6.35, flange Ø22, bolt circle Ø16	1	seller’s part (model in step/flange_coupler_6.35mm.step)

Slicer settings

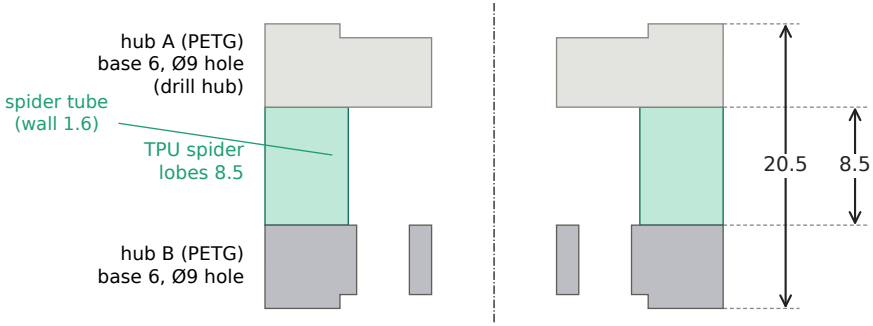
Setting	PETG Matte (plates, plots, rings, hubs)	TPU 85A (cushions, spider)
Feeding	AMS lite or external spool	external spool only, spool raised
Drying	yes (PETG Matte absorbs moisture)	70 °C for 8 h, then dry storage
Nozzle / bed	255 °C (230–260) / 70 °C (65–75)	225 °C (215 if it strings) / 35 °C, no glue
Layer / width / speed	0.2 / 0.42 mm / standard, fan 30–50 %	0.2 / 0.42 mm / 20 mm/s (15 first layer), fan 100 %
Support	none (hub B: under its recess; drill hub: under its Ø33 base)	none; retraction 0.8 mm at 25 mm/s; 2 mm³/s max
First print	one clip plate: it must click like sample 4	one cushion on a plot: firm friction fit

Jaw coupling — hubs A and B, TPU spider

Replaces the rigid drill → flange link. Hub A is the jaw end of the drill hub (page 16); hub B is screwed on the Ø22 flange (4 × M3 on Ø16).



Top view (jaw side of hub A, on the drill hub)



Assembled section, plane y = 0 (lobes 180° left, 0° right)

Files: DrillAdapter+hub (solidworks/) · hub_b_flange_side · spider_tpu · assembly_coupler

- Hub A (drill side): base Ø33 × 6, Ø22.2 × 1 centering recess, Ø9 centre hole, plain jaws with a 0.6 mm lead-in chamfer on their top flank edges.

  It is no longer screwed: it is printed in one part with the drill adapter V4, the drill hub (page 16).
- Hub B (flange side): base Ø33 × 6, Ø22.2 × 1 recess, 4 pockets Ø3.8 through the base for M3 × 4 × Ø4.2 inserts, Ø9 central hole for the shaft end;

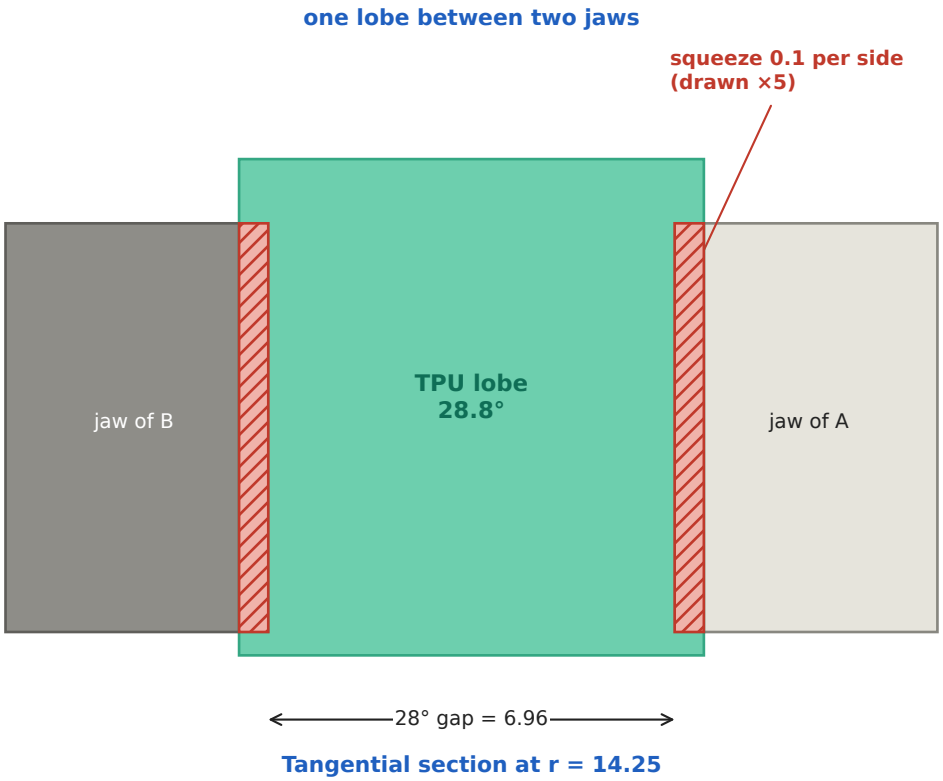
  flange screws M3 × 6, Ø5 × 3 heads on the flange side (0.5 mm clearance to the boss).
- Spider: TPU 85A (95A if it heats up), 100 % infill, a plain prism printed flat with the lobes upright: no overhang, no support, no small feature.

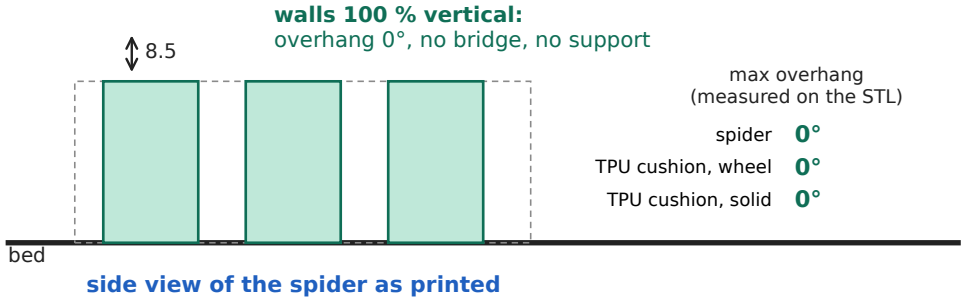
  It stays on the hubs by a light friction fit (next page): lobes 28.8° in the 28° gaps between the 32° jaws. Either face goes on either hub.
- Hub B and drill hub in PETG (or PA-CF), 4 perimeters; hub B base on the bed, supports only under its recess (drill hub: page 16).

  Allowable misalignment ≈ 0.3 mm / 1°; twist ≈ 1.5° (95A) to 4° (85A) at 3.3 N·m (≈ 2.7 MPa on the lobes).
- Order: drill hub onto the drill spindle, B onto the flange, spider onto B's jaws, then slide the motor in: the jaws engage. Keep a spare spider.

Spider fit — a plain prism held by a light squeeze

No snap, no peg, no overhang: printed flat, the spider stays on the hubs by friction between its lobes and the jaws. Dimensions in mm.



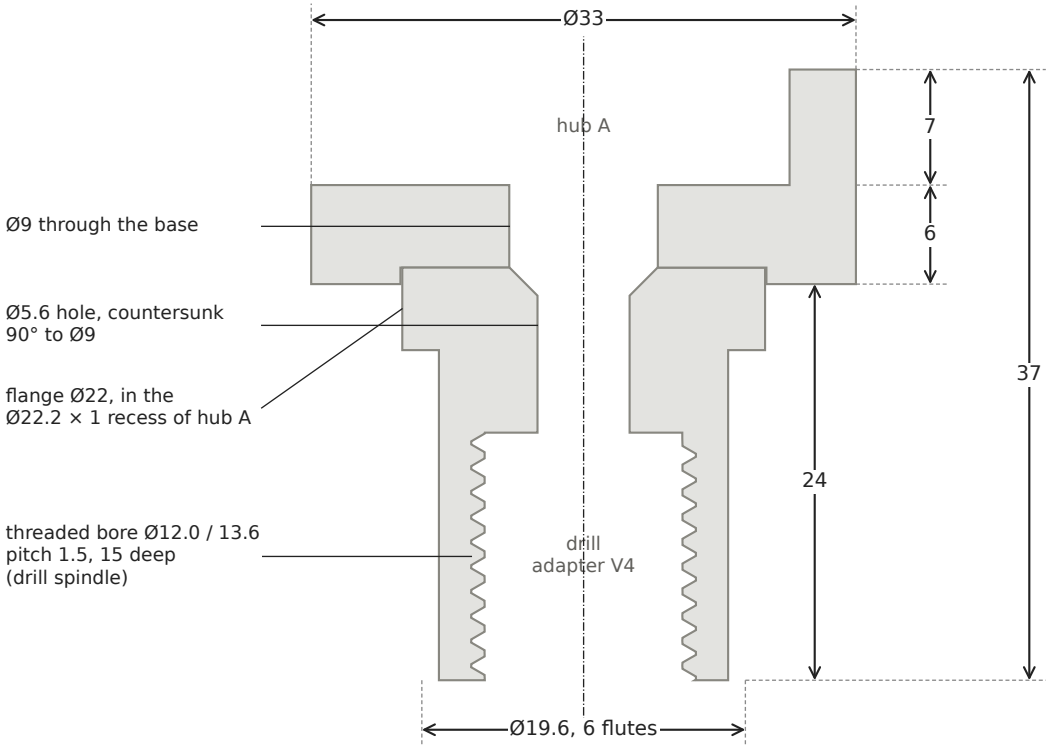


How it works

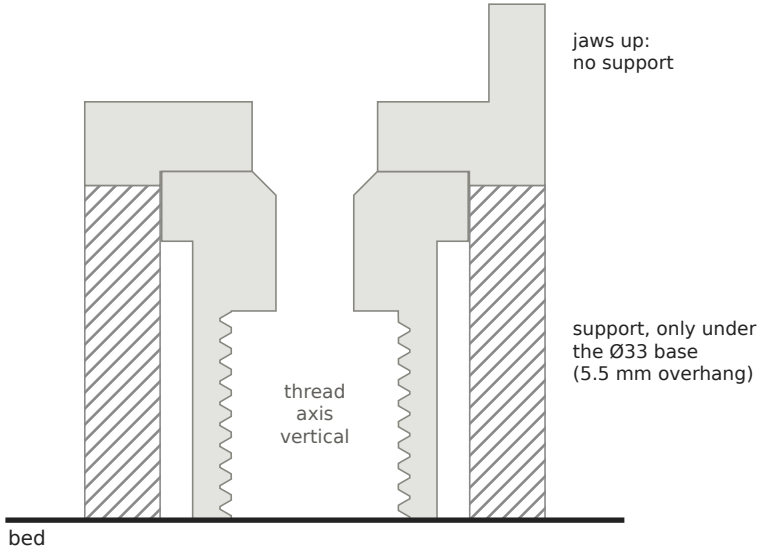
- The 6 lobes are 28.8° wide and the gaps between the jaws 28°: each lobe is squeezed by 0.1 mm per side. The spider is also captive between the two hubs.
- Friction (estimate, to be checked): about 5 N per flank, so ≈ 30 N holding on one hub and ≈ 30 N to push the other hub on. Nothing pulls on it in service.
- Nothing is printed that needs a support: no bead, no peg, no socket, no bridge. All the TPU parts have 0° of overhang.
- Tuning: the squeeze is the constant PRELOAD in source/coupling/coupling_parts.py (mm per side). Print 3 spiders at 0.0 / 0.1 / 0.2 and keep the one that grips firmly but can still be pressed on by hand. TPU printed 0.1 mm undersize gives no grip at all: raise PRELOAD.
- If it still slips out during assembly: a strip of double-sided tape on the end faces, or a drop of flexible adhesive on one hub only.
- Lead-in: 0.6 mm 45° chamfer on the two top flank edges of every jaw lets the lobes slide between the jaws.

Drill hub — drill adapter V4 and hub A in one part

Section through the axis of solidworks/DrillAdapter+hub.STL, made in SolidWorks from drillAdapterV4.SLDPRT and hub A. Dimensions in mm, measured on the STL.



Section through the axis and one jaw



Print: threaded end on the bed, jaws up

Drill hub: one printed part, no screws

- Drill adapter V4 (threaded bore for the drill spindle) and hub A, joined in SolidWorks: solidworks/DrillAdapter+hub.SLDASM (drillAdapterV4, hub_a_drill_side).
- Compared with the previous version: the 4 M3 × 8 screws and the 4 inserts between them are gone, and the jaws are centred on the bore by construction.
- Same jaws as before (3 × 32°, Ø25 → Ø33, 7 high, 0.6 mm lead-in): spider and hub B unchanged. Central Ø5.6 hole, reached through the Ø9 hole of the base.
- PETG, 4 perimeters, threaded end on the bed, support only under the Ø33 base. The STL holds two bodies (adapter, hub) that touch on the Ø22 face:

  keep them as one object in the slicer. First print: screw it onto the spindle by hand; it must go on without play and without forcing.

Hub B

- Flange screws M3 × 6 (2 mm flange + 4 mm insert); their Ø5 × 3 heads sit on its outer face, 0.5 mm clear of the Ø10 boss (head edge at r = 5.5).

Using the clip in your own SolidWorks supports

Files: clip_sample4_profile.dxf (outline), clip_cut_tool.step (solid), docs/clip_sample4_drawing.png. Origin on the plot axis, mouth toward +x.

Two ways

- DXF: open clip_sample4_profile.dxf in SolidWorks (File > Open, import as a 2D sketch), copy the sketch and paste it into a sketch on the face of your plate; place the origin on the plot axis. Cut-Extrude through the 6 mm of the plate.
- Solid: Insert > Part, clip_cut_tool.step (6 mm), origin on the plot axis, then Combine > Subtract with your plate body.

Rules to keep the validated behaviour

- Print the plate flat (page 13). The clip is only validated in that orientation.
- Keep the plate 6 thick and at least 15 wide along x, slot centred in x (7.5 from each edge) and the slit start at x = −0.5.
- Keep 1.6 mm or more of material outside the outer slit (slit outer edge at ±5.8 from the plot axis).
- Plot Ø6.0 exactly: the interference is 0.30 per side only with that diameter.

Coordinates from the plot axis (x toward the mouth, y across)

closed end R3.3 centred on the axis, straight walls y = ±3.3 from x = 0 to 3.5

inner ramp to y = ±2.7 at x = 4.5, lug flat x = 4.5 to 5.5 at y = ±2.7

entry ramp to y = ±3.6 at the plate edge x = 7.5

relief slits 1.0 wide, y = 4.8 to 5.8 and −4.8 to −5.8, from x = −0.5 (rounded end) through the edge

The six clips of the proof of concept

#	Tongue	Interf.	Mouth	Per clip*	4 clips*
1	0.8	0.30	5.4	≈ 8 N	≈ 32 N
2	1.0	0.30	5.4	≈ 16 N	≈ 63 N
3	1.2	0.30	5.4	≈ 27 N	≈ 109 N
4 (selected)	1.5	0.30	5.4	≈ 53 N	≈ 213 N
5	1.0	0.45	5.1	≈ 29 N	≈ 117 N
6	1.2	0.45	5.1	≈ 51 N	≈ 202 N

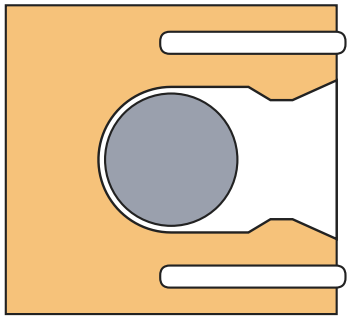
* Beam theory: tongue = 5 mm cantilever, flexural modulus 2 GPa, friction 0.3, two tongues per clip; real values are expected to differ by 50 % or more. The coupons of the proof of concept were printed to measure them.

If sample 4 is too stiff

- Reprint the plates with sample 3 (tongue 1.2) or 2 (tongue 1.0): change T_TONGUE and DELTA at the top of build_slide_mount.py, or the tongue and slit offsets in your sketch (slit y = 3.3 + tongue).
- The ring supports do not change.

If the tongues show white stress marks

- The root strain of sample 4 is ≈ 2.7 %. Stress marks after cycling mean the tongue is at its limit: go to sample 3.



Checks, validation plan and open items

What was verified in CAD, what has to be measured on the printed parts, and what is still unknown.

Verified in CAD (not printed yet)

- Every STL is watertight; the check assembly (2 supports, 4 plots, 8 cushions, plank, wood) has no interference other than the designed 0.1 mm friction fit of the cushions.
- Clearances: plate to cushions 0.3 (above, below), plate to plots 0.3 (around), head B to ring 0.8, plate end to wood 0.4, ring to wood 1.0.
- The clip is the sample 4 geometry of the proof of concept, generated from the same constants. The coupling files are those of the previous repository (same volumes); the drill hub joins hub A and the drill adapter V4 in SolidWorks.

Validation plan (in this order)

T1 Calipers: printed plot Ø6.0 ± 0.05, slot 6.6, mouth 5.4, cushion ID 5.8 on the shoulder. Drill hub screwed on the spindle by hand, without play.

T2 On the desk: measure H, the shaft axis to the plank underside (36.0 assumed) and set the cushion thicknesses (page 7).

T3 Bench: one support on two plots at pitch 56; 10 slide-ins; click, unlatching force with a scale along the axis.

T4 On the desk: turn the axis by hand; axis offset ≤ 0.3 mm, tilt ≤ 1°; run at low PWM and watch the offset under the couple (page 8).

T5 Cycles and creep: 50 insert/remove, then 48 h latched; look for white marks at the slit ends; re-measure.

T6 Vibration: ADXL345 on the plank and on the ring, with and without the cushions; compare with page 8 and keep the softest cushion that holds.

Open items

- Axial spacing L = 58.0 between the ring centres, from the 44.5 measured between the rings: check it with the drilling jig before drilling (page 11).
- Mouth direction: +x assumed toward the coupler; mirrored plate provided.
- Rings: taken from your SLDPRT, all original vertices within 0.002 mm. Open: the direction of the rear ring along x is as in your file (closed rim at x = 0). The key of the front ring is at the bottom right, as in your file.
- Plank ≥ 15 mm thick for the Ø3 × 25 screws, otherwise Ø3 × 20. H (36.0) sets the cushion thicknesses.
- Extraction force ≈ 4 × 53 N ≈ 210 N by calculation: to be measured; sample 3 is the fallback.
- Axial float: with clip sample 4 a plot can move about 3.7 mm between the slot end and the lugs. The coupling jaws (7 mm) accept it, but a motor that creeps back loses engagement; the spider's friction does not hold it. If it matters, move the lug 1.5 mm toward the slot end (a new clip to validate).
- Plate to pad joint: the couple (1.65 N·m per support) puts ≈ 80 N of tension on one M3 insert (estimate), with only 1.2 mm of ring wall under a pocket.
- Cushion stiffness, damping and the 0.3 mm plays are assumptions: page 8 is an estimate, the ADXL345 decides.

Assumptions

Plank underside 36 and wood face 27 from the SolidWorks sketches; plank 15–16 mm thick; M3 × 4 × Ø4.2 inserts in Ø3.8 pockets; screws Ø3 × 25 DIN 7997 type; torque 3.3 N·m shared by two supports.